🏥 Sehat-Assist AI

Pakistan's First Autonomous Multi-Agent Healthcare Assistant

Built on Qoder — A Complete End-to-End Medical Navigation & Safety Platform

🎯 Problem Statement

Every day in Pakistan, millions of people face critical healthcare roadblocks that cost time, money, and sometimes lives:

- Patients unknowingly consume fake or substandard medicines (Jali adviyat).
- Families receive complex lab reports in English medical jargon and panic, not understanding what it means.
- Patients waste hours standing in the wrong hospital department queues.
- People struggle to find the nearest specialized hospital in emergency situations — especially at night, when it's unclear which hospitals are even open.
- People often know a specialist by reputation but have no reliable way to find which hospital they currently practice at or when.
- Appointments are missed, and patients forget to take life-saving medications on time.

> **Sehat-Assist AI solves all of this through an autonomous, voice-first, multi-agent AI system — built to assist, never to diagnose.**

💡 What We Built

Sehat-Assist AI is a multi-agent AI ecosystem designed for the common Pakistani. It takes unstructured inputs (voice symptoms, images of medicines, lab reports, or prescriptions) and autonomously:

1. **Verifies** the authenticity of medicine boxes and flags anomalies for the patient to confirm — never issuing a consume/don't-consume directive itself.
2. **Translates & Simplifies** medical lab reports into local language voice notes without making medical diagnoses.
3. **Suggests a routing department** based on symptoms — explicitly labelled as a routing suggestion, not a diagnosis.
4. **Locates** the nearest appropriate hospital using real-time GPS and mapping APIs, including live open/closed status where available.
5. **Surfaces specialist availability** — which doctors are associated with which hospitals and their general practicing hours, so patients aren't left searching blindly for a specialist they've heard is good.
6. **Books** appointments via an AI voice call that clearly discloses it is an AI calling on the patient's behalf.
7. **Generates** a digital E-Parchi (appointment slip) for the patient.
8. **Schedules** automated reminders for medicine intake based on prescription scans.
9. **Guides** first-time users around the app itself through a scoped onboarding assistant — answering "what does this app do" and "where is this feature," never medical questions.

🤖 Agent Architecture

```
USER INPUT
(Voice Symptoms / Camera Scans: Meds, Reports, Prescriptions)
                  |
                  v
        +-------------------------+
        |   MAIN ORCHESTRATOR      |
        |   Intent Detection       |
        |   Consent & Permission   |
        |   Management              |
        +-------------------------+
          |      |      |      |      |
          v      v      v      v      v
     +--------+ +------+ +--------+ +-----------+ +-----------+
     | PHARMA | |LINGO | | TRIAGE | | CARE-SYNC | | ONBOARDING|
     | CHECK  | |-MED  | | AGENT  | |   AGENT   | | / HELP    |
     +--------+ +------+ +--------+ +-----------+ | (RAG BOT) |
                   |                               +-----------+
                   |
                   v
              +-----------------+
              | GEO-LOCATOR +    |
              | AVAILABILITY     |
              | LAYER            |
              +-----------------+
                   |
                   v
           +-----------------+
           | AUTO-BOOK &      |
           | E-PARCHI         |
           | (AI self-        |
           | discloses on     |
           | call)            |
           +-----------------+
                   |
                   v
        +-------------------------+
        | EMERGENCY ESCALATION     |
        | LAYER                    |
        | -> routes red-flag cases |
        | to local emergency line  |
        +-------------------------+
                   |
                   v
          FINAL RESPONSE & ACTION TO USER

     [ PRIVACY & CONSENT LAYER wraps every agent ]
```

🧠 Agent Descriptions

1. Main Orchestrator Agent

Role: The Traffic Controller

Pain: patients don't know which "mode" they need — they just have a symptom, a pill, or a report, and no idea where to start.

- Requests device permissions (Microphone, Camera, Location) on first launch, with a plain-language explanation of why each is needed.
- Analyzes the user's input and routes it to the correct specialized agent.
- Maintains session context and logs consent state.

2. Pharma-Check AI (Fake Medicine Flagging)

Role: The Quality Inspector

Pain: a family buying medicine at a small pharmacy has no way to know if it's genuine before their child swallows it.

- Activates the camera to scan medicine packaging (barcode / 2D DataMatrix, aligned with DRAP's national serialization mandate).
- Extracts batch and registration data and checks for visible anomalies.
- Surfaces a flag for the user to act on rather than instructing them medically.


```
{ "scanned_item": "Panadol CF", "drap_number": "012345",
  "authenticity_status": "COULD NOT BE VERIFIED",
  "reasoning": "Batch number format mismatch.",
```

"recommended_action": "Please confirm with your pharmacist before use.",
"disclaimer": "Sehat-Assist AI assists — it does not make medical decisions for you." }

3. Lingo-Med AI (Report Simplifier)

Role: The Friendly Guide

Pain: a mother gets her son's lab report back in English medical jargon and has no one to explain it to her before the next appointment.

- Processes uploaded lab reports using OCR.
- Identifies out-of-range parameters in plain language.
- Generates a simplified explanation via TTS in the local language.
- **Strict Guardrail:** appends the "Assist, not Diagnose" disclaimer on every single output.

4. Smart Triage AI

Role: The Reception Desk

Pain: patients waste hours standing in the wrong department queue because no one told them where to go first.

- Listens to patient symptoms in natural language.
- Suggests the likely relevant department (e.g., Cardiology, Gastroenterology) as a routing suggestion only, never a diagnosis.
- Flags emergency keywords and hands off directly to the Emergency Escalation Layer.

5. Geo-Med Locator & Availability Layer

Role: The Navigator

Pain: in an emergency at night, people don't know which nearby hospital is actually open — and separately, people often know a specialist by reputation but have no reliable way to find which hospital they currently practice at or when.

- Fetches real-time GPS coordinates (with explicit user consent) and queries a mapping API for nearby hospitals with the identified department.
- Surfaces live open/closed status for hospitals where this data is available from the mapping provider.
- For hospitals and clinics without reliable live status, displays a clearly labelled prototype/seed dataset with a "please confirm by calling ahead" notice — never presented as verified real-time fact.
- Surfaces specialist-to-hospital and general practicing-hours information, sourced from a curated seed dataset for the prototype stage, with a roadmap toward integration with existing verified doctor networks in Pakistan for production use.
- Returns distance, estimated travel time, and a navigation link.

6. Auto-Booking & E-Parchi Agent

Role: The Executive Assistant

Pain: patients — especially the elderly or those uncomfortable on the phone — struggle to call and negotiate an appointment slot themselves.

- Opens every call by disclosing it is an AI assistant calling on the patient's behalf.
- Conducts a real-time AI voice conversation to request an appointment slot.
- Generates a downloadable "E-Parchi."
- In the prototype phase, calls route only to designated testing numbers — never to real hospital lines.

```
SEHAT-ASSIST AI - E-PARCHI -------------------------------- Patient Name: Ali Raza Date: 21 August 2026 Time: 6:00 PM ------------------------------ Hospital: XYZ Clinic Department: Gastroenterology Status: CONFIRMED (Voice) -------------------------------- Distance: 2.5 km from you Show this pass at counter
```

7. Care-Sync AI (Prescription Reminder)

Role: The Caretaker

Pain: patients, especially the elderly, forget to take medicines on time and no one tracks it for them.

- Reads doctor prescriptions (parchi).
- Extracts medicine name, dosage, and timings.
- Schedules automated push notifications or reminders.

8. Emergency Escalation Layer

Role: The Safety Net

Pain: a genuine emergency needs a real next step, not just an in-app alert.

- Triggered by Smart Triage's emergency keyword flag.

- Surfaces the relevant emergency contact number for the user's area.
- Never delays or replaces calling for real help — it's an accelerant, not a gatekeeper.

9. Onboarding / Help Assistant (RAG Bot)

Role: The Guide

Pain: a first-time user, especially someone less tech-savvy or elderly, opens the app and doesn't know what it does or where any feature is — leading them to abandon it before ever using it.

- Answers plain-language questions about the app itself — what each feature does, where to find it, how to use it — retrieved from the app's own documentation and feature descriptions (RAG over Sehat-Assist AI's own content).
- Strictly scoped to product/app questions only — it does not answer medical questions.
- Any message that looks like a symptom, health concern, or medical question is redirected to the Orchestrator for routing to Triage or the relevant medical agent, rather than being answered directly.
- **Boundary guardrail:** this agent is intentionally not open-ended. It is scoped strictly to onboarding and in-app guidance, and must not attempt to engage with or answer anything outside that scope.

10. Privacy & Consent Layer

Role: The Guardrail

Pain: this app touches voice, camera, GPS, and lab data — some of the most sensitive data a person owns.

- Explicit, per-feature consent screens for microphone, camera, and location.
- Data encrypted at rest; processed on-device where feasible.
- Clear retention policy communicated to the user.
- Health data is never sold or shared with third parties.

🔁 Complete Flow Example (Triage to Booking)

```
[18:01:00] ORCHESTRATOR: Intent = SYMPTOM_TO_HOSPITAL
[18:01:01] TRIAGE AGENT: Symptoms analyzed. Suggested department = Gastroenterology
           (routing suggestion, not a diagnosis). Urgency = HIGH.
[18:01:02] GEO-LOCATOR: GPS = Gulshan-e-Iqbal. Querying Maps API...
[18:01:03] GEO-LOCATOR: Nearest facility = Patel Hospital (1.2 km), open now.
[18:01:03] ORCHESTRATOR: "Aap ke nazdeek Patel Hospital hai. Appointment milaun?"
[18:01:08] USER: "Haan bhai mila do."
[18:01:09] AUTO-BOOKING AGENT: Initiating Twilio Voice Call to verified testing number.
[18:01:10] TWILIO LOG: AI: "Assalam-o-Alaikum, main Sehat-Assist AI hoon, Ali Raza
           ki taraf se appointment ke liye baat kar raha hoon."
[18:01:12] TWILIO LOG: AI: "Unhe pait dard ki appointment chahiye."
[18:01:15] TWILIO LOG: Receptionist: "6 baje aajaen."
[18:01:16] TWILIO LOG: AI: "Shukriya, confirm." Call disconnected.
[18:01:17] AUTO-BOOKING AGENT: E-Parchi generated for 6:00 PM.
[18:01:18] ORCHESTRATOR: UI updated with E-Parchi.
```

👥 Team Task Division Plan

Track A: Vision & Analytics Engine	Track B: Action & Navigation Engine
Focus: Camera, OCR, Image Processing, Cron Jobs	**Focus:** Voice, Telephony, GPS, Mapping
Pharma-Check AI (Medicine Scanning) Lingo-Med AI (Report OCR & Simplify) Care-Sync AI (Prescription Parsing)	Smart Triage AI (Voice to Department) Geo-Med Locator & Availability Layer (GPS, Maps API, Doctor/Hospital Hours) Auto-Booking Agent (Twilio Voice API)
APIs: Vision API, TTS, Schedulers	**APIs:** Speech-to-Text, Twilio, Google Maps

Cross-cutting responsibilities (shared across both tracks): Privacy & Consent Layer, Emergency Escalation Layer, and the Onboarding/Help Assistant (RAG Bot over the app's own documentation).

📌 Market Timing & Differentiation

Pakistan's Drug Regulatory Authority has set October 9, 2026 as the deadline for mandatory 2D barcode and serialization on all human and veterinary medicine packaging, enabling consumers to verify authenticity and expiry by smartphone scan — timing that lines up closely with this build.

Source: Sustainable Packaging Middle East & Africa, reporting on DRAP's Pharma Track and Trace System mandate.

Existing Pakistani platforms such as Marham and Oladoc already offer large-scale doctor directories and appointment booking. Sehat-Assist AI's differentiation is not the directory itself, but surfacing hospital and specialist availability automatically inside an emergency triage-and-navigation flow — a context those planned-booking platforms are not built for. Similarly, at least one existing product (VerifyPharma) offers AI-powered fake-medicine detection alone; our differentiation is the full chain from detection through triage, navigation, AI-assisted booking, and reminders in one voice-first flow.

⚠ **Important — read before demo day:**

1. Sehat-Assist AI operates strictly on an "Assist, Not Diagnose" policy across every agent.

2. Simulated bookings in the prototype phase route only to Twilio-verified testing numbers, never to real hospital lines.

3. The Auto-Booking Agent discloses it is an AI on every call.

4. Hospital and specialist availability data in the prototype is a curated seed dataset for demo purposes where live data isn't available, clearly labelled as such, with real live data used wherever the mapping provider supplies it.

5. Regional-language TTS coverage beyond Urdu is a roadmap item, not a confirmed production capability.